Circular motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate inward net force to a change in velocity direction.

SUCCESS CRITERIA - I can...

1. Draw tangent velocity and inward acceleration.
2. Use v^2/r or a supplied acceleration.
3. Identify the real interaction providing centripetal force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “centripetal” and “radius” to explain your answer. Sentence starters are optional.

centripetal: directed toward the center of a circular path

radius: distance from the center to the edge of a circle or sphere

net force: the vector sum of all forces on one object

01 Warm-up: bring back an earlier idea

Two balls launch horizontally from the same height at **3** and **15 m/s**. Ignore air resistance. **Compare fall times** and horizontal ranges.

Word help: horizontal = along a level line, usually the x direction

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

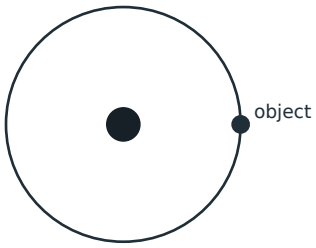
02 Worked example: explain the prediction

Reference: $a_c = v^2/r$; $F_{net} = mv^2/r$.
c means centripetal: toward the center. This is the net inward force, not an extra force.

Worked reasoning: A **2 kg** object moves at **3 m/s** in a circle of radius **3 m**. Inward acceleration is v^2/r=**3 m/s²**; net force is **6 N** inward.

Your turn: Draw tangent velocity and inward net force at two different points.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Circular motion

antocciphysics

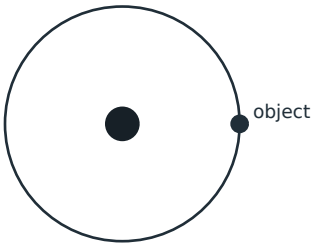
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and explain the diagram or equation

1. **Add velocity** and acceleration arrows to the marked point on the circle.
2. **Name one interaction** that could provide the inward force.

Sentence starter: In my work, I used … to show … . I labeled it … .



The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: 1 2 3

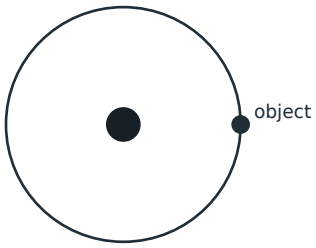
04 Apply the physics

A **2 kg** small classroom car follows a circular track. Its inward acceleration is **3 m/s²**.

1. Draw and **label the inward net force** and **find its magnitude**.
2. How would that force change if inward acceleration doubled?

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

Circular motion

antocciphysics

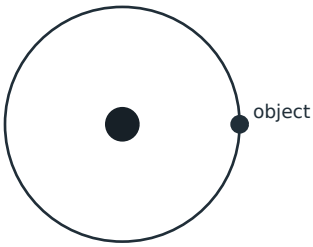
Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A string breaks while a ball moves in a circle. Neglect other forces over the next instant. **Sketch its initial path after** release.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **8 kg** cart starts from rest under **80 N** net force right for **7 s**. **Find acceleration** and final velocity. **State one thing you** are treating as true.
Label the given and predicted quantities on the diagram.

Sentence starter: I predict … . I used … to make this prediction because … .

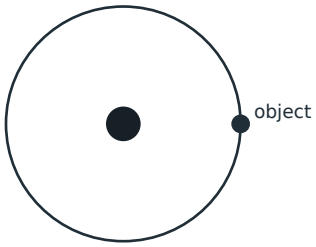


Marzano check - circle: 1 2 3

07 Exit check

A car turns at steady speed. Can net force be zero?
1. Explain using velocity.
2. **Label the radius** and **show velocity** and inward acceleration on the circle.

Sentence starter: I think … . I can point to …, which shows …, so … .

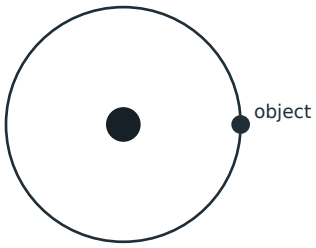


Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**. At the same radius and mass, compare forces for speeds **3** and **6 m/s**. Why is the answer not just twice as much? **Label the radius** and **show velocity** and inward acceleration on the circle.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3